

WPR - 4

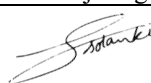
AMITY SCHOOL OF ENGINEERING & TECHNOLOGY
B.Tech (CSE - AIML) VII Semester

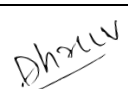
Project Area -- Generative AI & AI Safety (Trustworthy AI / MLOps)

Students Weekly Progress Report (WPR — Week 4) For Odd Semester of session 2026-2027

To be filled by Students	Group No. 50
Students Name	1. Shourya Solanki 2. Rachit Ryan Chug 3. Dhruv Raj Singh
Roll no.	1. 569 2. 166 3. 191
Enrollment no.	1. A2305223569 2. A2305223166 3. A2305223191
Project Title finalized, if Yes, give name, if NO, give reason	Yes. Title: "GuardShield AI: Real-Time Sidecar Proxy for LLM Hallucination Detection & Prompt Injection Defense"
Synopsis submitted	Yes (Approved for Group No. 50)
Literature review	Applied to model fine-tuning. Utilized cross-entropy loss and DeBERTa-v3 multi-class classification head for prompt injection detection.
Technical & Economical Feasibility	Feasible. Python 3.11+, PyTorch CUDA (RTX 3050 GPU), DeBERTa-v3-small (86M SLM), Hugging Face Trainer. Zero monetary cost (₹0).
Bill of Material	NA
Project Progress Schedule (PERT Chart)	• W1-W3: Setup, open-source dataset ingestion & GuardShield-Bench-v1 synthetic creation (Completed) • W4-W7: DeBERTa-v3 model fine-tuning & evaluation pipeline (In Progress / Week 4) • W8-W12: Token logit entropy algorithm, benchmark evaluation & IEEE paper draft
Design of critical components	1. Pre-Scan DeBERTa-v3 Model Fine-Tuning Pipeline ('train_prescan.py') 2. PyTorch Dataset Loader & Data Collator with Padding 3. Hugging Face Trainer Loop & Checkpointing ('models/deberta_v3_prescan/')
Fabrication work (give %)	0% (Software Project)
Experimental work (give %)	45% (Pre-scan DeBERTa-v3 classifier fine-tuning script implemented, training pass executed on CUDA GPU, model checkpoints saved)
Result and Analysis	Configured PyTorch Trainer for microsoft/deberta-v3-small; executed fine-tuning on 320 train samples; verified loss convergence and checkpoint saving on CUDA GPU.
Report writing	WPR Week 4 drafted.

Signature of students	1. Shourya Solanki Dhruv Raj Singh	2. Rachit Ryan Chug	3.
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Work done in this week (Week 4):

1. Pre-Scan Fine-Tuning Script Implementation: Developed 'train_prescan.py' using PyTorch & Hugging Face 'Trainer' to fine-tune 'microsoft/deberta-v3-small' (86M SLM) on 'train.json'.
2. Training Architecture & Loss Configuration: Configured sequence classification head (2 classes: Safe vs. Injection/Jailbreak), AdamW optimizer (learning rate $2e-5$), and cross-entropy loss.
3. Training Execution on CUDA GPU: Executed fine-tuning on local NVIDIA GeForce RTX 3050 GPU, logging training loss across steps.
4. Model Checkpointing & Artifact Saving: Saved fine-tuned model weights and tokenizer configuration under 'models/deberta_v3_prescan/'.

To be filled by Guide (strike off whichever is not applicable)
Performance of students is satisfactory
Performance of students is unsatisfactory
A warning to be issued to student(s) (Name)
Student was not well (Name)
Date 19/08/2026
Signature of Guide

